

Project Nexus: Product Requirements Document (PRD)

Project Vision: To dismantle the "Attention Economy" by replacing centralized, AI-driven silos with a distributed, encrypted framework. Nexus returns control of the social graph and digital assets to the user while providing a sustainable income model for the independent infrastructure providers that power the network.

1. Target Audience

- **The "Ad-Fatigued":** Users exhausted by intrusive marketing and the commodification of their personal data.
- **The "Signal-Seekers":** People tired of "noise," rage-bait, and unsolicited interactions from "randos."
- **The "Human-First" Community Individuals:** Wary of AI-generated content flooding their feeds and seeking authentic, verified human interaction.
- **The Small Provider:** Hobbyists or small-scale entrepreneurs looking to monetize spare hardware/bandwidth through Home Servers and Relays.

2. Core Objectives

- **Algorithm Autonomy:** Move from "Platform-Controlled AI" to "User-Defined Curation." Users choose which plugins/schemas filter their data, not a black-box engagement algorithm.
- **Noise Reduction:** Implement economic and cryptographic "gates" (Attention Fees/PRE approvals) to ensure that digital interactions are intentional and high-quality.
- **Decentralized Income:** Enable a "Service Layer" where users can earn Nexus Credits by providing storage, relaying encrypted traffic, or mirroring content for their social circles.
- **Invisible Privacy:** Deliver a world-class UX where high-level encryption (Proxy Re-Encryption) happens silently in the background.

3. Functional Requirements

3.1 Distributed Identity & Auth

- **DIDs (Decentralized Identifiers):** Users own their identity keys. No central database of users exists.
- **Approval Handshakes:** All private data sharing requires a bi-directional "vouch" or "approval," which triggers the generation of kfrags for secure access.

3.2 Modular App Ecosystem

- **Nexus Drive (V1 App):** A distributed "Dropbox" clone. Sharded file storage with PRE-based folder sharing and real-time sync across personal nodes.

- **Nexus Feed (V1 App):** A "Twitter" clone focused on "Signal over Noise." No global trending based on ads; feeds are generated strictly from the user's trusted graph and "approved" sources.

3.3 The "Home Server" (Node Utility)

- **Persistence-as-a-Service:** An optional, lightweight server (Raspberry Pi/Old PC) that seeds the user's data 24/7.
- **Earning Potential:** Home servers can be configured to act as "Public Relays" for the broader network, earning a share of protocol fees for their uptime.

3.4 Monetization & The "Attention Gate"

- **The Junk Folder Toll:** Unsolicited messages land in a "Paid Inbox." Senders must pay a user-defined fee to notify the recipient.
- **One-Way Payment Bridge:** Crypto-to-Nexus Credit conversion to facilitate tipping, subscription-based "Apps" (e.g., Nexus OnlyFans), and infrastructure payments.

4. User Experience (UX)

- **The "Nexus Shell":** A modular desktop and mobile application built on Tauri 2.0.
- **Contextual Discovery:** Use QR codes and "Magic Links" shared through existing channels to bootstrap the social graph without a global search directory.
- **Curation Plugins:** A marketplace of open-source "Feed Filters" that allow users to manage how they see data (e.g., "Chronological Only," "Media Only," "Vouched Only").

5. Security & Legal Compliance

- **Non-Provider Architecture:** By design, the developers cannot intercept, decrypt, or moderate content, placing the app outside the "Service Provider" definition of Bill C-22 (Canada).
- **Atomic Transactions:** Payments for content (kfrag release) and message "Unlocks" are handled by local smart-contract logic to prevent fraud.

Feature	The Problem	The Nexus Solution
Monetization	Selling user data to advertisers.	Infrastructure fees and P2P "Attention Gates."
Algorithm	AI-driven engagement loops.	User-owned, modular curation plugins.
Spam	Free, infinite bot accounts.	Economic cost for unsolicited messages.

Feature	The Problem	The Nexus Solution
Hosting	Giant, vulnerable data centers.	Distributed "Home Servers" and Relays.